Florian Pothin

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Quantitative Economist – Climate, Trade & Macro – Ph.D. Candidate in Economics – Université de Rennes – CREM UMR6211

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Defence expected: November 2026 – available from 2027

Profile

Quantitative economist working on climate change and international trade, and building theoretical models of how the economy and the climate shape each other. A large part of the work is about measurement, because the carbon numbers this field relies on are partly chosen: my research measures how far those choices move reported emissions, how they travel from firms through trade into national statistics, and what they mean for climate policy built on reported data. I also work extensively with artificial intelligence, and on it: machine learning is part of my toolkit, and the environmental footprint of AI services is one of my research objects. Developed through a CIFRE partnership with Toovalu, where I now work as an independent consultant; every paper ships with a public replication package.

Education

Ph.D. in Economics — *Université de Rennes – CREM UMR6211* — 2023 – Nov. 2026 (expected)

"The Decarbonization Gap: Firms, Nations, and the Economics of an Unfinished Transition"

Supervisors: Fabien Rondeau & Tovonony Razafindrabe | CIFRE industrial partnership: Toovalu (ANRT no. 2022/1627)

M.Sc. Financial and Economic Engineering — *Université de Rennes* — 2018 – 2020

Graduated with honors. Focus: macroeconomics, econometrics, data science, financial modelling, risk management.

B.Sc. Economics — *Université de La Réunion* — 2015 – 2018

Valedictorian. Focus: macroeconomics, microeconomics, mathematics, statistics, data science.

Professional Experience

Data Scientist – Research & Impact — *Toovalu, Nantes (CIFRE, now independent consultant)* — Apr. 2022 – Present

- Built econometric, forecasting and machine-learning models, with their data pipelines, embedded in a B2B SaaS platform for corporate climate strategy, ESG reporting and GHG-emissions analytics.
- Developed a firm-, sector- and country-level emissions observatory integrating CDP, ADEME, Bloomberg, EXIOBASE and macro-environmental indicators; benchmarking runs on corrected rather than reported numbers.
- Applied-AI lead: designed and shipped the platform's production LLM systems (retrieval-augmented generation, structured extraction) and its life-cycle-assessment module; built models for emissions estimation and reporting-bias correction.

Quantitative Research Intern — *Natixis Investment Managers, Paris* — Mar. 2020 – Jan. 2021

- Return-prediction models for a multi-asset fund (Python, R); ESG integration in portfolio construction and performance attribution.

Research Assistant – Macroeconomic & Country Analysis — *Université de Rennes – CREM* — Apr. – Aug. 2019

- Country-risk and macroeconomic analyses for OPEC member states; Bloomberg and World Bank data; applied research reports.

Research

Journal names are given once papers are accepted. Replication packages are public; drafts on request.

- **Voluntary Climate Pledges Widen What Firms Measure, Not What They Abate [Job Market Paper]**
  Sole-authored. Under review. Replication: 10.5281/zenodo.21627772.
  After a voluntary pledge, firms report 2.5 more Scope-3 categories within four years and reported Scope 3 rises 0.3–0.5 log points through the wider boundary alone; the rise disappears once each firm's boundary is held fixed. Scope 1 shows a precisely estimated null.
- **Beyond Reported Emissions: Uncovering Hidden Corporate Carbon Footprints**
  With F. Rondeau, T. Razafindrabe and A. Macherey. Under review, second round; revision submitted July 2026.
  Diagnoses omission and under-reporting for each of the 15 GHG-Protocol Scope-3 categories, then estimates the missing part: value-weighted, reported footprints miss 21.7% [19.5, 24.2] of the corrected total, two thirds of it in use of sold products.
- **A Product-Level Trade-LCA Framework for Embodied Carbon Accounting in Global Trade**
  Sole-authored. Under review since June 2026. Preprint: Research Square rs-10627706; code: 10.5281/zenodo.20719018.
  Links bilateral HS6 trade to life-cycle emission factors (about 5,000 products, 120 countries, 1995–2023), with LLM-assisted matching audited blind twice (17 then 8 wrong per 100). World embodied carbon in traded goods: 14,368 $MtCO_2e$ in 2023; intensity varies thirteen-fold within the median sector.
- **Transparency and Verification in the Economics of Decarbonization**
  Sole-authored. Under review. Replication: 10.5281/zenodo.21719291.
  Codes 2,578 studies on eight research fronts: 6.5% of disclosure studies analyse Scope 3, two thirds of embodied-carbon studies do not name their input–output database, and 95.7% of decoupling studies never assess Paris compliance.
- **Trade Shocks Redraw the Global Carbon Map but Do Not Shrink It**
  Sole-authored. Ready for submission. Replication: 10.5281/zenodo.21889661.
  Product-level embodied carbon through China's WTO accession, the US–China tariff war and the 2022 sanctions on Russia, generalised over 35 accessions and 9 sanction episodes: integration pulls carbon in slowly (+31% at ten years), sanctions push it out fast (up to −88%), and the carbon relocates rather than shrinking.
- **Is the Environmental Kuznets Curve Identified?**
  Sole-authored. Submission in preparation.

A turning point counts as identified only if its bootstrap interval fits strictly inside the observed income support. On 105 countries (1995–2023), no specification passes for production- or consumption-based emissions, at 99–100% simulated power; the trade-adjusted measure passes at the threshold of the criterion.

- **Sectoral Climate Legislation and Corporate Targets**
  Sole-authored. Thesis chapter; not currently under submission.
  On 1,958 listed firms in 28 countries (2015–2023), each additional sectoral climate law raises the probability of an SBTi commitment by 1.66 percentage points; the emissions test is underpowered in available data, and the paper says so.
- **Multi-Criteria Environmental Assessment of Generative AI Services: a Life-Cycle-Based Method**
  With T. Gilormini. Working paper, v1.0, August 2026; Zenodo deposit forthcoming.
  Assesses AI services over their full life cycle (training, inference, hosting, networks, devices) on four simultaneous impact categories; reconstructs cloud emission factors (the electricity-mix convention alone shifts results by roughly a factor of four in carbon-intensive regions) and replaces generic device figures with primary measurements per user-journey step.
- **Fair Shares for Firms: What the Flat Rate Hides**
  In preparation.
  Against alternative fairness rules for the same carbon budget, the flat 4.2%-a-year rate under-charges 89% of committed Scope-1+2 tonnage, concentrated in heavy industry.
- **Endogenous Abatement as Green Capital in a Heterogeneous-Firm General Equilibrium**
  With T. Razafindrabe. In preparation.
  Hopenhayn-style general equilibrium where abatement is the green share of capital: of a 42% emission cut, 86% is reallocation across firms and 14% within-firm greening; permit design decides whether the price bites.

Funded Projects & Service

- **ANR I-DAVE** *(2024–2027)* – AI and knowledge-based framework for life-cycle-assessment reliability, with Centrale Nantes (LS2N), UTC (Roberval), Toovalu and Cetim. Grant held by Toovalu; I carry out its work on the company's side.
- **IRIS-E**, Université de Rennes – appointed member of the axis-2 working group ("Interdisciplinary and co-constructed research"), two-year mandate from December 2023.
- **SBTi Expert Working Group** – member, Data Quality Assurance and Claims, Science Based Targets initiative.

Distinctions

- **Prix Transition**, Trophées Valorisation 2024, Campus Innovation, Université de Rennes, with T. Razafindrabe and F. Rondeau, for the CIFRE CREM–Toovalu project.
- **Winner, "Prédiction de sinistres" data challenge** (Generali), 2018 edition, with M. Riou; presented at the data science seminar of Stéphane Mallat's chair, Collège de France.

Conferences & Seminars

- **International Economies: Challenges Ahead**, Ryukoku University / Yokohama National University / Sciences Po Aix / Aix-Marseille School of Economics, Kyoto (Feb.–Mar. 2026) [presentation]
- **Research workshop**, Shiga University, Japan (2026) [presentation]
- **11th International Macroeconomics Workshop**: International Macroeconomic Issues of Climate Change, CREM, Université de Rennes (2025) [presentation]
- **SESTEF 2024**, Paris (Dec. 2024) [presentation]
- **FFM29**, 29th Forecasting Financial Markets Conference, IPAG, Oxford (Sep. 2024) [presentation]

Technical Skills

Programming	Python, R, Matlab, Dynare, SQL
Econometrics & Macro	Panel data, event studies and staggered designs, randomization inference, time series (VAR/SVAR), GMM, IV/2SLS, hurdle models; heterogeneous-agent GE modelling, calibration, forecasting
Machine & Deep Learning	XGBoost, Random Forest, Elastic-Net, meta-learning; PyTorch, TensorFlow; GNN (PyTorch Geometric), CNN, RNN; LLM-assisted measurement with audited reliability
Data	Bloomberg, CDP, ADEME, EXIOBASE, BACI, World Bank, OECD.Stat; API integration, data pipelines, multi-source harmonization
Practices	Reproducible research, public replication packages, Git, HPC, LaTeX

Teaching Experience

Université de Rennes, M.Sc. Ingénierie Économique et Financière (IEF) 2023 – 2025 | ≈100h ETD
Macrodynamics: DSGE and real business cycle models | Macroeconometrics: time series

Business schools 2022 – 2026 | ≈150h total
Supervised Machine Learning in R, IA School M.Sc. (2024–2026); Python & AI Frameworks, EPSI (2022–2025); Mathematics for Finance, IGEFI (2024–2026); Green Finance, IET (2022–2023).

Languages & Interests

French (native) | English (professional working proficiency; research and teaching) | German (A2) | Japanese (A1)
Interests: macroeconomics, financial markets, ecological transition, AI for economic intelligence, open data, hiking, music.

References

Available upon request: **Fabien Rondeau** (thesis supervisor, Université de Rennes – CREM) | **Tovonony Razafindrabe** (co-supervisor, Université de Rennes – CREM)